

Abstract

Abstract I

How should you choose the judges, jurors, or reviewers who form a panel — and does that upstream choice change how well you can evaluate them *without an answer key*? A panel can be selected many ways — by competence, by a representative lottery (**sortition**), by ideological bloc, or at random — and, separately, its noisy judgments can be evaluated blind: given the agreement/disagreement pattern among three binary judges, the `ntqr` package's error-independent (EIE) evaluator returns logically consistent estimates of item prevalence and per-judge accuracy with no labels at all. But that evaluator takes the panel as given. We join the two questions and ask whether the *rule that forms the panel* changes the oracle-referenced error of the no-answer-key evaluation — how far the blind estimate lands from the answer-key result, lower being better.

Abstract II

On a fully deterministic instrument (96 seeds, 96 experts, 300 items), the dominant lever is *which* rule forms the panel, not its size: competence-first selection recovers best (0.037), while representative, single-bloc, and random selection collapse together — *by construction*, because with independent judge errors composition cannot move an estimator that only sees agreement. Supplying the missing channel — same-group judges sharing a latent, marginal-accuracy-preserving error confound — makes the strategies fan out monotonically as within-bloc coupling rises: representative sortition stays flat while single-bloc selection degrades, the gap widening from 0.000 to 0.112. Within this instrument the relationship is closed-form: recovery error tracks the panel's **Herfindahl concentration index** over the axis a shared error rides on — minimized exactly by a balanced (representative) draw, maximized by a single bloc — and a continuous representativeness dial confirms error rises monotonically with it.

Abstract III

The protection is **conditional**: re-keying the confound to an axis the lottery does not balance erases the protection (0.147→0.229). The lesson for selecting and evaluating panels is thus a falsifiable, simulation-bounded prediction, not a preference for any one rule — representativeness protects blind recovery precisely when the panel balances the attribute a shared error rides on. Evidence is synthetic and oracle-scored; in a single small live model (gemma3:4b) the synthetically-best competence-first rule was the worst, illustrating that a selection rule validated on parameterized judges need not carry over to prompted ones — a hypothesis to test, not an established caution. All methods and documentation are openly available at the public repository **docxology/ntqr_allotment**.